Prediction of Plant Vigor and Final Biomass Using Early Phenotypic Traits

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Overview

1. Exploratory Data Analysis
2. Predictive Modeling
3. Interpretation
4. Summary

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1. Exploratory Data Analysis

1.1 Data Description

- **Plants**: Poplar (*Populus*) grown in ORNL's Advanced Plant Phenotyping Lab
- **Imaging period**: ~35 days of side-view imaging
- **Data shape**: 3408 rows x 62 columns

Category	Example Variables	Description
Metadata	barcode, genotype, analyse date, etc.	Plant identity, genotype info, and imaging timeline
Morphological Features	area, convex_hull_area, height, perimeter, solidity, etc.	Shape and size measurements from side-view images.
Spectral Features	Red_mean, blue_mean, green_mean, hue_mean, etc.	Statistics derived from RGB, HSV, and LAB opponent color spaces

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1.2 Data Cleaning & Merging

- Standardized plant identifiers from Plant Info and barcode files.
- Merged barcode metadata with imagined-derived feature tables
- Removed rows with missing or mismatched identifiers
- Ensured numeric types and consistent formatting across morphology & spectral fields.

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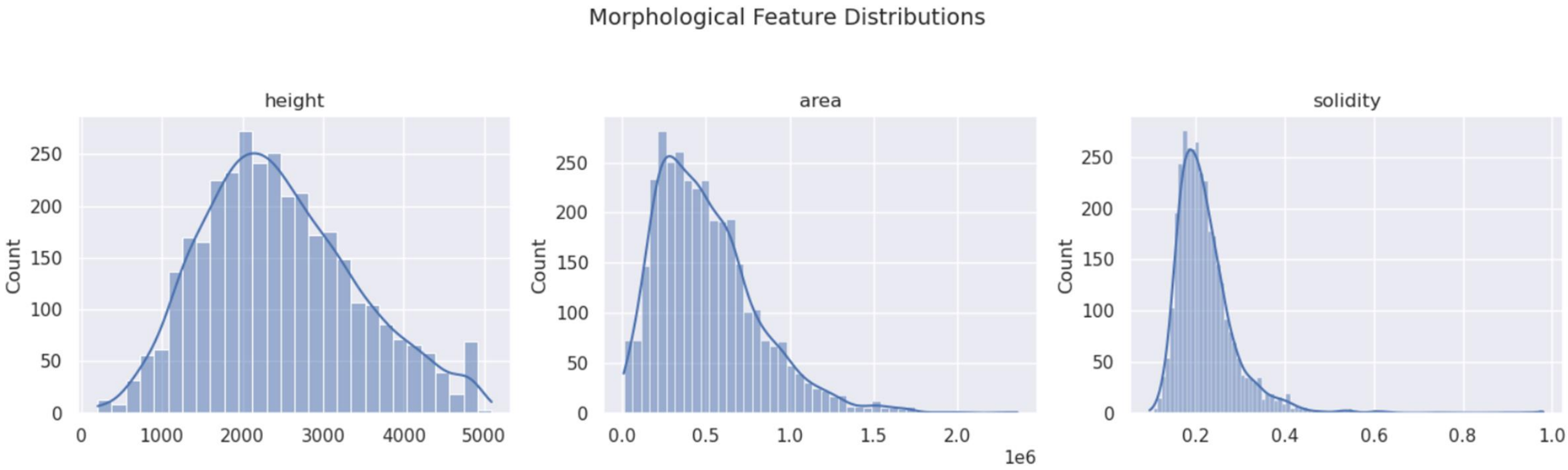
1.3 Morphological Explorations - Distributions

- Height shows a roughly normal distribution.
- Area and solidity are right-skewed reflecting plant size variability.



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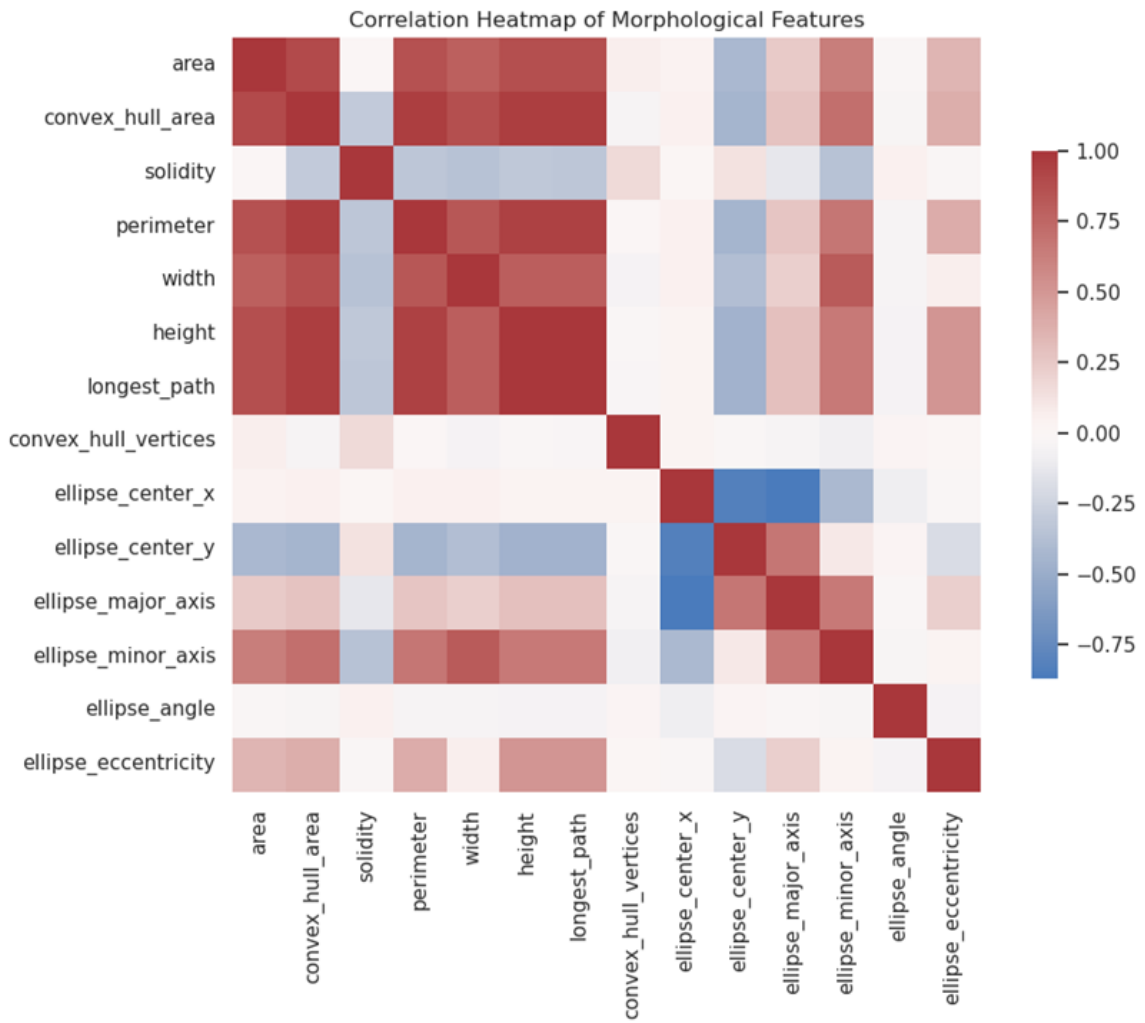
1.3 Morpholoigcal Explorations - Correlations



- Consistent relationships appear among key size traits reflecting expected biological coupling.
  - e.g . Height, area, and width

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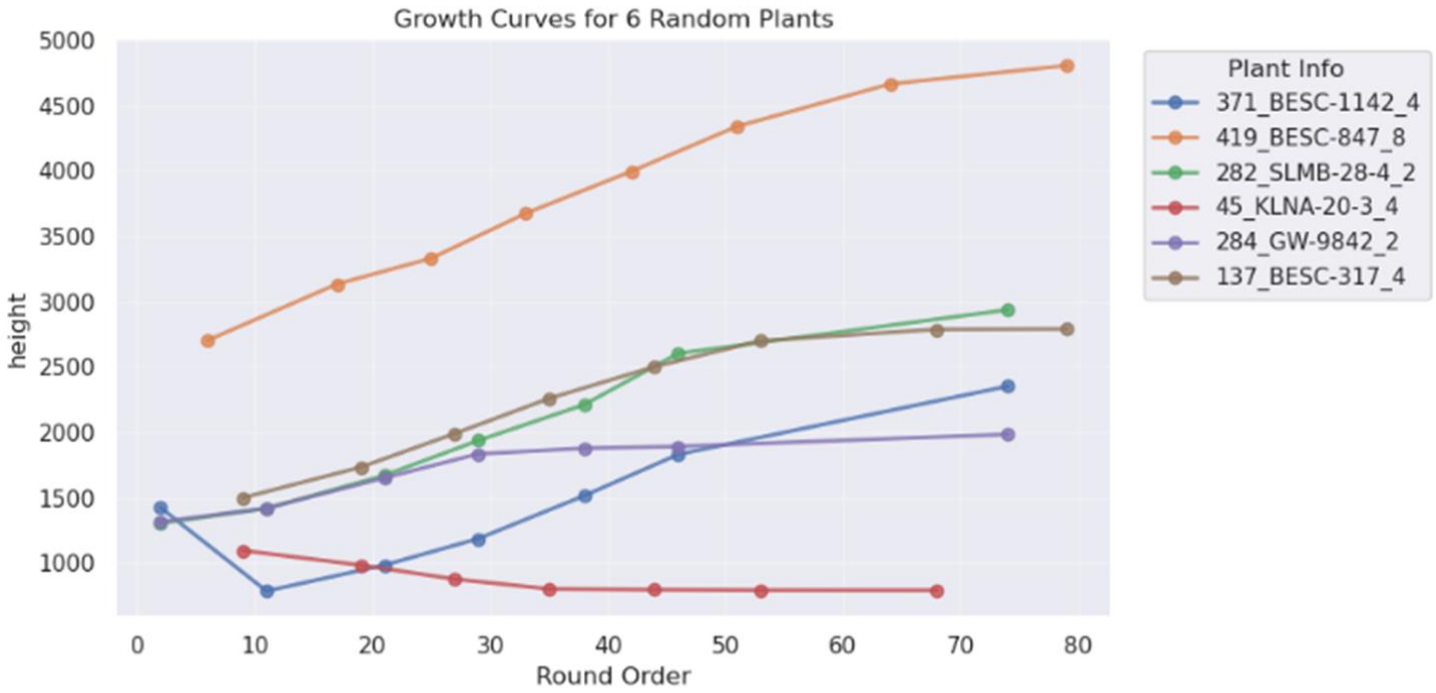
1.4 Growth Curve Structures – Individual Level

- Randomly selected individual plants show diverse growth trajectories over ~35 days of imaging.
- Some plants grow steadily; others plateau or fluctuate due to biological or imaging noise.
- Individual-level curves help quantify growth variability.



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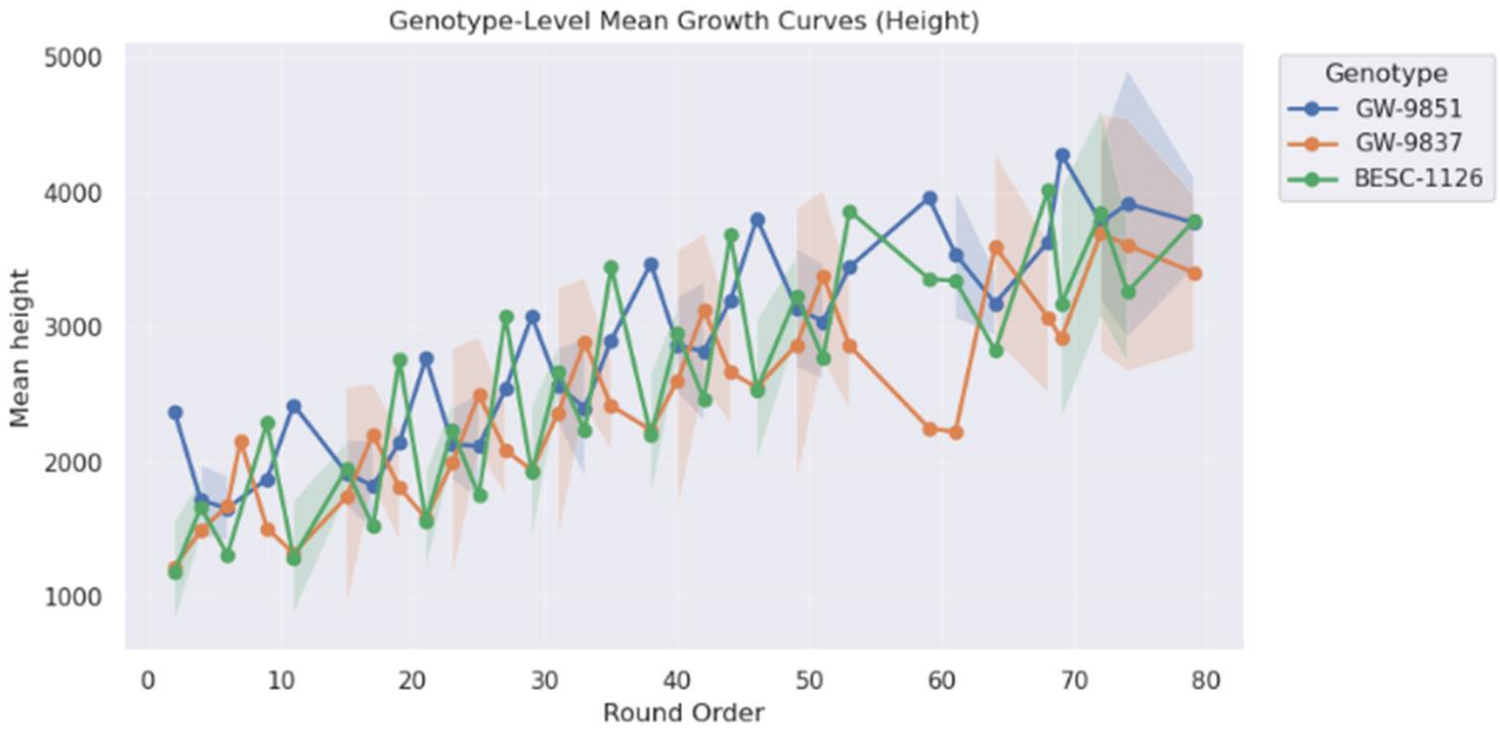
1.4 Growth Curve Structures – Genotype Level



- Genotypes exhibit characteristic growth patterns and slope differences.
- Mean curves reveal genetic influence on height trajectories.
- Standard deviation bands show within-genotype variability.

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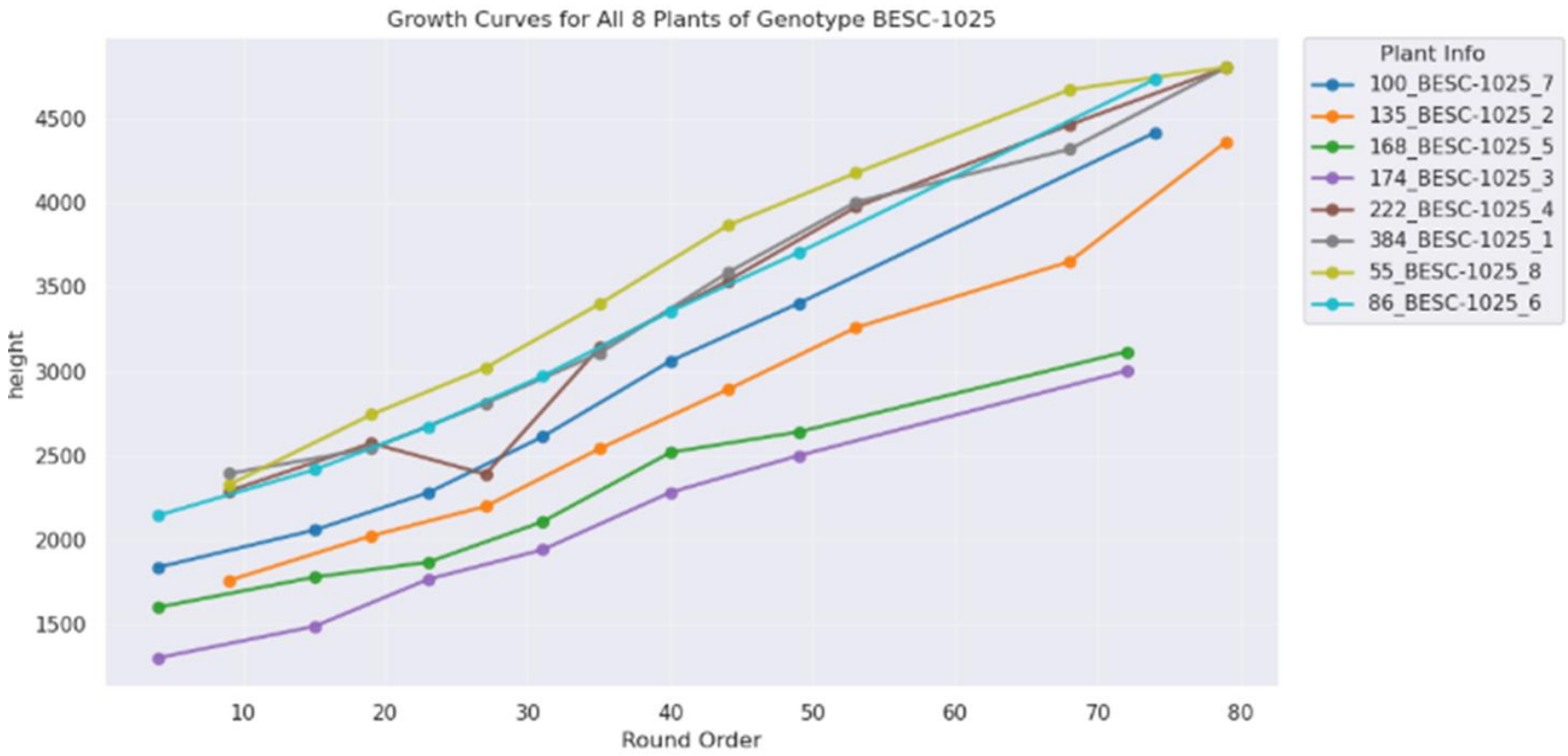
1.4 Growth Curve Structures – Looking at BESC-1025



- 75% (6 of 8)of the BESC-1025 clones were flourishers.
- All BESC-1025 plants show strong upward growth patterns.
- Clear trajectory consistency suggests genetic stability within this line.

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2. Predictive Modeling

2.1 Overview of Modeling Projects

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2.2 Early Spectral Signals Overview

Goal:

- Identify early spectral signals that predict long-term plant performance

Approach:

- Early-round spectral modeling
- Δ-spectral comparisons
- SHAP feature interpretation

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2.2 Defining and Identifying Flourishers

Flourisher – top 25% in maximum height and area

Dataset: 446 plants
- 364 non-flourishers
- 82 Flourishers

Of the 82 flourishers:
- 53 were of the BESC variety
- 16 were GW
- 3 were CHWK
- 3 were KLNA
- 2 were of the CA Variety
- 1 was CHWH
- 1 was KLND
- 1 was LILC
- 1 was NHTB
- 1 was SLMB

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2.2 Most Efficient Genotypes

The genotypes that had the best flourish percentage were:

BESC-1025 (75%)
BESC-1068 (67%)
BESC-1095 (62.5%)
BESC-1098 (62.5%)
BESC-856 (62.5%)
KLNA-20-3 (60%)
BESC-97 (50%)

=== Flourishers by Genotype ===

genotype	total_plants	flourishers	flourish_rate
BESC-1025	8	6	0.750
BESC-1068	6	4	0.667
BESC-1095	8	5	0.625
BESC-1098	8	5	0.625
BESC-856	8	5	0.625
KLNA-20-3	5	3	0.600
BESC-97	8	4	0.500
BESC-117	8	3	0.375
BESC-304	8	3	0.375
GW-7994	8	3	0.375
GW-9851	8	3	0.375
CHWK-27-4	8	3	0.375
BESC-430	8	3	0.375
GW-9919	8	3	0.375
BESC-45	8	2	0.250
BESC-354	8	2	0.250
GW-9837	8	2	0.250
GW-9823	8	2	0.250
CA-01-03	8	2	0.250
GW-10986	8	2	0.250
BESC-1154	8	2	0.250
BESC-1108	8	2	0.250
KLND-20-1	5	1	0.200
BESC-1017	5	1	0.200
NHTB-27-1	5	1	0.200
BESC-1041	6	1	0.167
BESC-1126	8	1	0.125
BESC-1143	8	1	0.125
BESC-1146	8	1	0.125
SLMB-28-4	8	1	0.125
GW-9806	8	1	0.125
LILC-26-2	8	1	0.125
BESC-284	8	1	0.125
CHWH-27-4	8	1	0.125
BESC-847	8	1	0.125

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2.2 Comparing the different models

Models Tested:
- Structural & Spectral combined model
- Δ - spectral temporal model
- Early-round spectral model
- SHAP Analysis

Results:
- The Structural & Spectral combined was a good overall model, but we wanted to see if we could predict solely using spectral features
- The Early-round spectral model (left) performed better than the Δ-spectral temporal model (right)

Classification report (test set):

	precision	recall	f1-score	support
0	0.857	1.000	0.923	18
1	1.000	0.400	0.571	5
accuracy			0.870	23
macro avg	0.929	0.700	0.747	23
weighted avg	0.888	0.870	0.847	23

Classification report (test set):

	precision	recall	f1-score	support
0	0.810	0.944	0.872	18
1	0.500	0.200	0.286	5
accuracy			0.783	23
macro avg	0.655	0.572	0.579	23
weighted avg	0.742	0.783	0.744	23

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2.2 SHAP Results

SHAP results:
Top features consistently linked to vigor:

- Red min (anthocyanin/chlorophyll balance)
- Blue yellow min (pigment stability & leaf health)
- Green red mean (chlorophyll concentration)
- Lightness/value (canopy reflectance & energy status)

Top 15 SHAP features overall (first round):

	feature	mean_abs_shap_overall
0	red_min	0.036550
1	blue_yellow_min	0.022020
2	green_red_mean	0.021910
3	hue_mean	0.018654
4	value_max	0.016607
5	saturation_mean	0.016232
6	red_max	0.015970
7	value_mean	0.015064
8	blue_max	0.015024
9	blue_yellow_mean	0.013625
10	lightness_max	0.013427
11	red_mean	0.012480
12	blue_mean	0.007821
13	blue_yellow_max	0.007701
14	lightness_mean	0.007067

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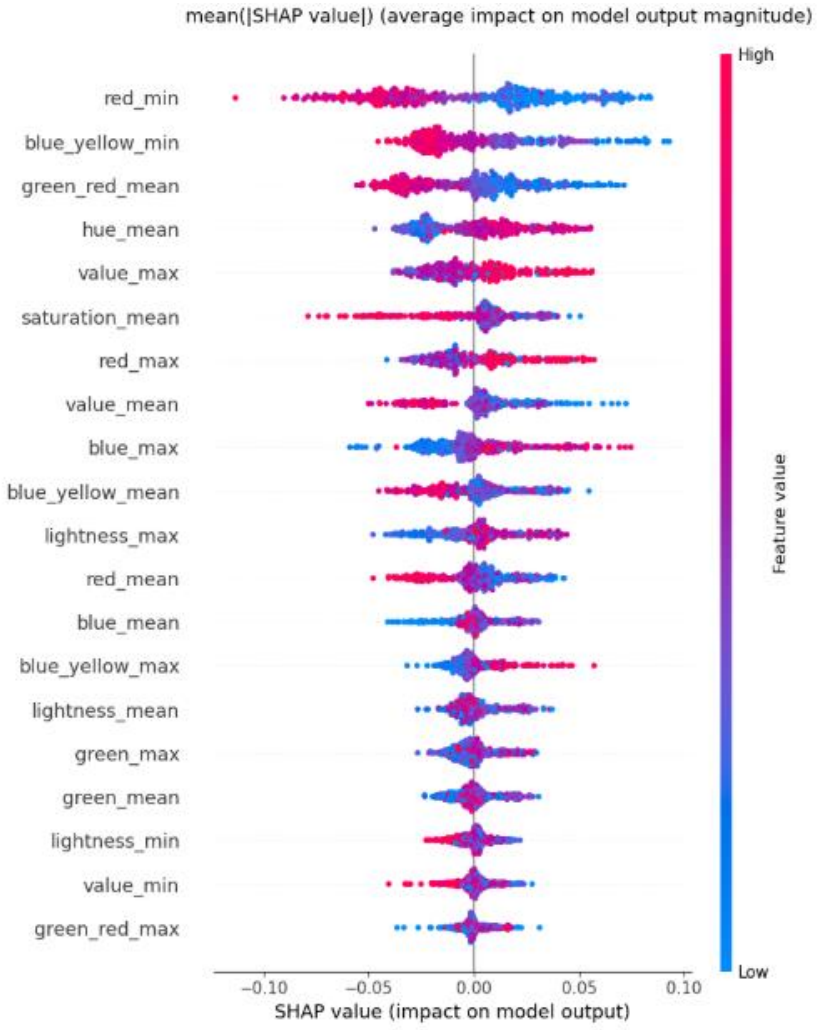
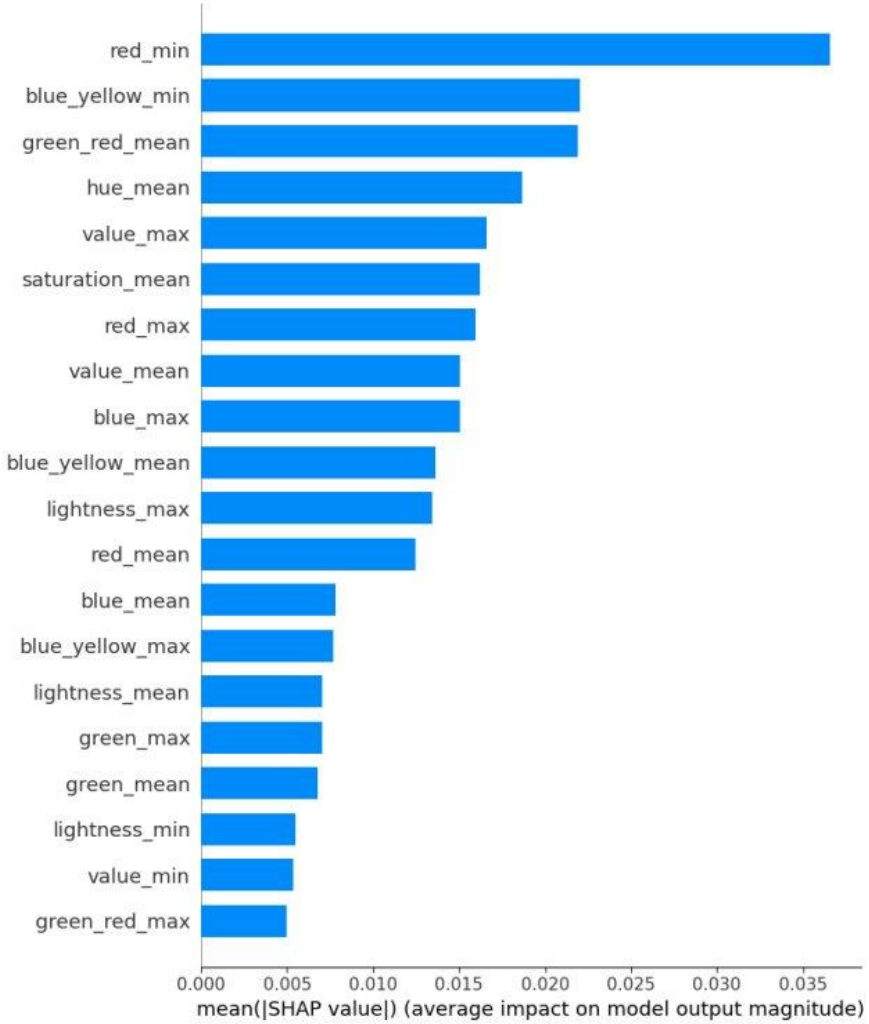
2.2 More SHAP Results





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2.2 Comparing SHAP values between Flourishers and Non-Flourishers

Top 15 features where SHAP importance differs between flourishers and non-flourishers:

	feature	mean_abs_shap_flourish	mean_abs_shap_not_flourish	diff_flourish_minus_not	abs_diff
10	red_min	0.052784	0.032893	0.019891	0.019891
22	blue_yellow_min	0.037856	0.018452	0.019404	0.019404
20	blue_yellow_mean	0.027382	0.010526	0.016856	0.016856
3	blue_max	0.028746	0.011933	0.016813	0.016813
15	lightness_max	0.026831	0.010408	0.016423	0.016423
24	hue_mean	0.032044	0.015637	0.016407	0.016407
33	value_max	0.029936	0.013604	0.016332	0.016332
11	red_max	0.028649	0.013114	0.015535	0.015535
30	value_mean	0.027653	0.012228	0.015425	0.015425
16	green_red_mean	0.034033	0.019179	0.014853	0.014853
8	red_mean	0.023682	0.009956	0.013726	0.013726
27	saturation_mean	0.027326	0.013733	0.013593	0.013593
12	lightness_mean	0.016662	0.004906	0.011756	0.011756
0	blue_mean	0.017086	0.005733	0.011353	0.011353
7	green_max	0.015674	0.005113	0.010561	0.010561

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2.2 Implications

Practical Implications:

- Early detection supports resource allocation
- Enables proactive selection of strong performers
- Identifies slow-growers for intervention or removal

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2.3 Predicting Final Biomass from Early Growth- Overview

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2.3 Predicting Final Biomass from Early Growth

- Goal: use the first 14 days of plant imaging to predict final side-view area (biomass proxy).
- Target: Final_Area = area at each plant's last recorded day.
- Question: "*Can we accurately predict final biomass using only early growth patterns?*

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2.3 Predicting Final Biomass from Early Growth – Early Window Feature Engineering

- For each plant, we summarize the first 14 days morphological growth features
- Growth rates:
  - $\text{area_slope} = \Sigma(x_i - \bar{x})(\text{area}_i - \bar{\text{area}}) / \Sigma(x_i - \bar{x})^2$
    - Growth rate of area (pixels/day)
  - height_slope → Growth rate of height (pixels/day)
- Size:
  - area_mean, height_mean→ Average size and height over 14 days
- Variability:
  - area_std, height_std → Growth variability (fluctuation in area/height)
- Last early measurement:
  - area_last, height_last → Last recorded area and height (Day 14 value)
- Shape Descriptor:
  - solidity_mean → Average shape compactness
- Color features from 14 days:
  - green_mean14, red_mean14, blue_mean14

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2.3 Predicting Final Biomass from Early Growth – Early Window Feature Engineering

- One hot encoding
  - Added *Plant.Genotype* to the feature table for each plant using one-hot encoding
  - Each represents genetic variation in growth response
  - Dropped one reference genotype to keep features compact

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2.3 Predicting Final Biomass from Early Growth – Plant-Level Modeling Table

- Each row = one plant, one early-growth feature row per plant, used as model input.
- Columns include:
  - 12 engineered early-growth features (slopes, means, stds, last values etc.)
  - 59 one-hot encoded genotype indicators
  - Target Final_Area (final biomass proxy)
- Train/test considerations:
  - groups = Plant.Info used for grouped splits
  - No plant appears in both training and testing within a fold.

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2.3 Predicting Final Biomass from Early Growth - Model Candidates and Grouped CV by Plant

- Tested 5 models
  - Lasso (alpha=0.0005, max_iter=20000)
  - Ridge (alpha=1.0)
  - Random Forest (n_estimators=500, min_samples_leaf=3)
  - Gradient Boosting(n_estimators=100, learning_rate=0.1, max_depth=3)

- Evaluation setup:
  - 5-fold GroupKFold, grouped by Plant.Info
  - Metrics: R^2, RMSE per fold, then averaged
  - This prevents leakage: plants in the test fold are never seen in training.

- Output:
  - results_df summarizing mean R^2 ± SD and mean RMSE ± SD across models

Model	R^2 (mean)	R^2 (sd)	RMSE (mean)	RMSE (sd)
Lasso Regression	0.739797	0.058224	177571.31	17533.51
Ridge Regression	0.739685	0.060162	177529.30	18000.76
Gradient Boosting Regressor	0.693693	0.058589	193215.19	16111.81
Random Forest	0.591773	0.051201	223779.05	11018.29

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2.3 Predicting Final Biomass from Early Growth – Models Performance

Final area is highly predictable from early morphological traits.

- Linear models (best performing)
  - Lasso: $R^2 \approx 0.74$, RMSE ≈ 178k pixels
  - Ridge: $R^2 \approx 0.74$, RMSE ≈ 178k pixels
  - Both models show low SD across folds (≈0.06), indicating stability.
- Tree-based models
  - Gradient Boosting Regressor: $R^2 \approx 0.69$ (RMSE ≈ 193k pixels)
  - Random Forest: $R^2 \approx 0.59$ (RMSE ≈ 224k pixels)
  - Less consistent than linear models.
- Conclusion:
  - Linear models (lasso/ridge) fit the data structure best.
  - Early morphological traits carry strong predictive information.

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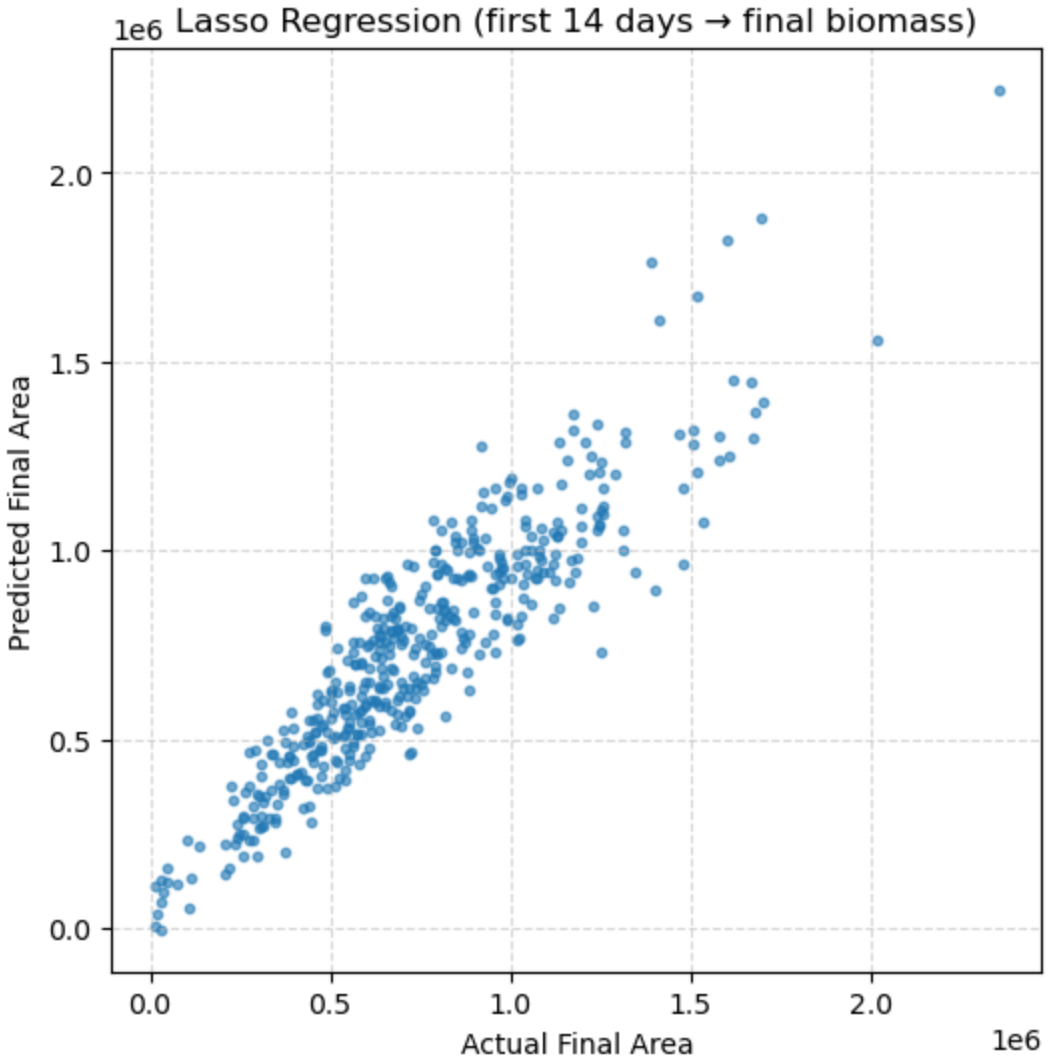
2.3 Predicting Final Biomass from Early Growth



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2.3 Predicting Final Biomass from Early Growth – Predicted vs Actual Area

Plant info	Predicted final area	Actual final area	Final day
31_BESC-1068_4	2,221,165	2,356,271	32
295_BESC-1095_1	1,880,824	1,695,388	31
441_BESC-304_8	1,821,582	1,597,736	32
27_BESC-1098_6	1,761,652	1,390,660	32
37_BESC-1068_1	1,676,599	1,515,358	32
121_BESC-1068_3	1,608,625	1,412,487	31
68_GW-10986_2	1,555,167	2,015,457	31
36_BESC-1146_7	1,449,047	1,619,619	32
87_GW-9837_4	1,444,879	1,664,851	32
78_GW-9837_5	1,394,972	1,698,347	31
419_BESC-847_8	1,366,564	1,677,349	35
152_BESC-1095_8	1,359,729	1,172,481	35
86_BESC-1025_6	1,332,630	1,237,251	32
328_BESC-856_3	1,321,880	1,503,498	35
175_KLNA-20-3_2	1,321,399	1,174,494	31

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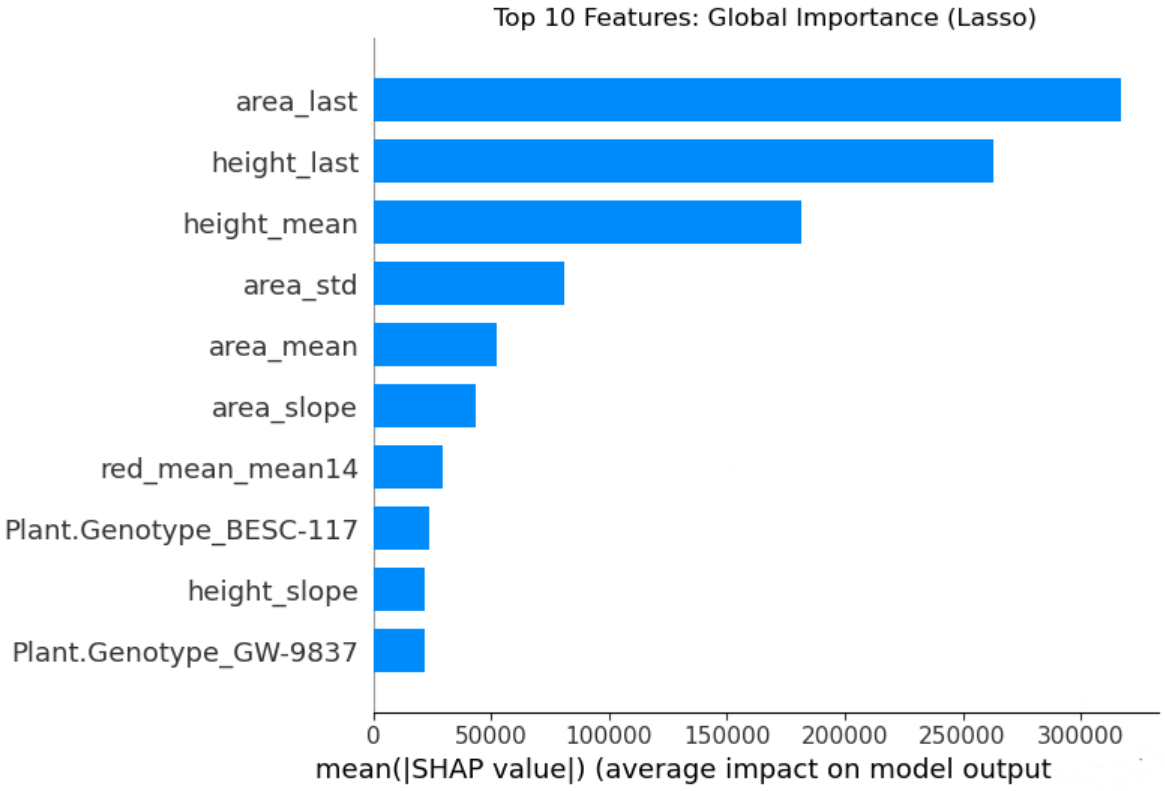
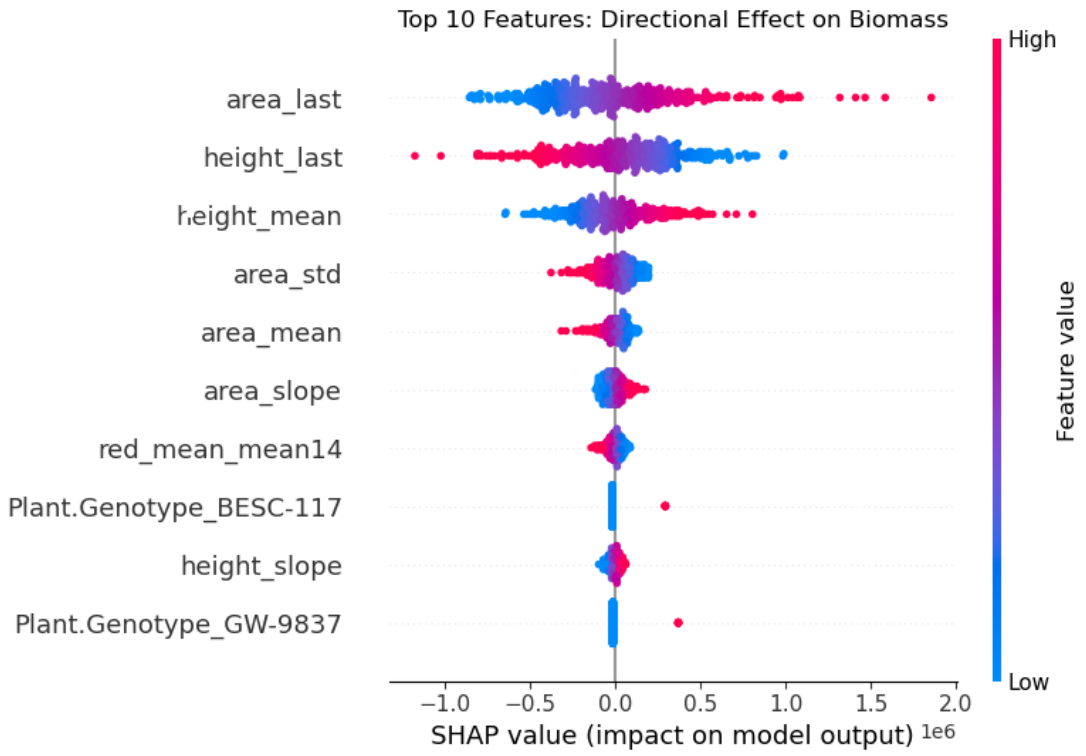
2.3 Predicting Final Biomass from Early Growth – Features Importance Using SHAP





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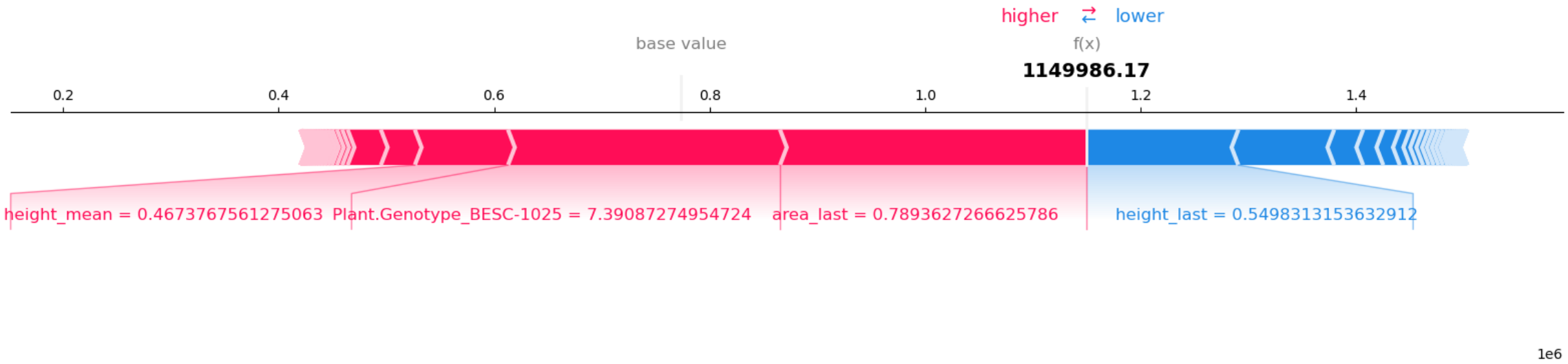
2.3 Predicting Final Biomass from Early Growth – SHAP Force Plot



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2.3 Predicting Final Biomass from Early Growth - Hold-Out Evaluation (30% Test Split by Plant)

- **Method**: Used *GroupShuffleSplit* rather than random row splitting.
- **Grouping Variable**: *Plant.Info* (Unique Plant ID).
- Ensures that all measurements from a specific plant reside exclusively in either the Training set or the Testing set.
- **Training Set (70%)**: 311 Plants
- **Test Set (30%)**: 134 Plants

Hold-out R^2	**0.736**
Hold-out RMSE	**176387.4**

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2.3 Predicting Final Biomass from Early Growth - StratifiedGroupKFold

- **Method**: Used *StratifiedGroupKFold* (k=5) rather than simple random CV.
- **Grouping Variable**: *Plant.Info* (Unique Plant ID), ensures no data leakage i.e. same plant never in both train/test.
- **Stratified**: *Genotype*, ensures every fold has a representative mix of all genotypes.
- Ensures that all measurements from a specific plant reside exclusively in either the Training set or the Testing set.
- **Per Fold**: Model trains on ≈ 356 plants (80%) and tests on ≈ 89 plants (20%). Repeated 5 times so every single plant

Fold 1	90 test plants
Fold 2	90 test plants
Fold 3	89 test plants
Fold 4	88 test plants
Fold 5	88 test plants

Hold-out R^2	0.749 (±0.031)
Hold-out RMSE	176387.4 (±11,290)

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2.3 Predicting Final Biomass from Early Growth - Leave-One-Genotype-Out (LOGO-CV)

- **Method**: Used Leave-One-Genotype-Out Cross-Validation (LOGO-CV) via an iterative "Round-Robin" approach with Ridge Regression.
- **Grouping Variable**: Plant.Genotype (Unique Genetic ID), ensures strict biological exclusion so the target genetics are never in the training set.
- **LOGO-CV**: Ensures that the model predicts biomass for a completely new genotype it has never seen before (testing genetic generalization).
- **Per Fold**: Model trains on all other genotypes ("General Population") and tests on the single target genotype. Repeated for every genotype to identify biological outliers.
- **Filtering**: Genotypes with less than 3 plants were excluded to ensure statistical validity.

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2.3 Predicting Final Biomass from Early Growth - Leave-One-Genotype-Out (LOGO-CV)

- Average LOGO-CV R^2 (Ridge): -0.371

Genotype	n_train	n_test	R^2	RMSE
KLND-20-1	440	5	0.871	123642.79
LILC-26-2	437	8	0.854	90294.21
BESC-1103	438	7	0.841	96497.01
BESC-1231	438	7	0.839	152347.80
KLNA-20-3	440	5	0.824	221453.37

Top 5 Genotype Easiest to Predicted

Genotype	n_train	n_test	R^2	RMSE
BESC-856	437	8	-2.981	435584.24
BESC-1166	437	8	-3.257	112571.11
BESC-1043	437	8	-3.449	175956.91
BESC-1032	440	5	-4.404	144267.63
GW-9842	437	8	-7.114	227589.60

Top 5 Genotype Hardest to Predict

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2.3 Predicting Final Biomass from Early Growth - Implications

Implications:

- Resource allocation using early growth rate morphological data.
- Selecting of strong performers bypassing months of wait time for final yield.
- Identifying plants based on unique growth deviations (outliers), flagging rare genetics for further study.

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3. Interpretation

3.1 Interpretation – Early Spectral Modeling

- Flourishers had significantly lower Red_min values on average (~59% lower than non-flourishers).
- BlueYellow_min was consistently among the top SHAP-ranked features across all spectral models.
- Early-round spectral model achieved 87% accuracy using only first-round features.

Interpretation:
→ Plants "signal" their future performance early, through pigment and reflectance patterns. For example, plants with low Red_min (higher chlorophyll & anthocyanin balance) were far more likely to become flourishers.

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3.2 Interpretation – Predicting Final Biomass from Early Growth

- Linear models (Lasso/Ridge) achieved **$R^2 \approx 0.74 \pm 0.06$**, outperforming tree-based models
- Final biomass (final area) is mostly determined by early growth rate and early size.
  - The top predictors included: area_last, area_mean, area_std, area_slope height_last, height_mean

Interpretation:
→ Early structural growth gives a strong, measurable preview of total biomass potential. The dominance of linear models and slope features indicates that the system behaves in a mostly linear biological range.

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3.2 Interpretation – How Well the Models Generalize

- GroupKFold: **$R^2 \approx 0.74$ with SD ≈ 0.06**, showing strong model stability
- StratifiedGroupKFold ensured genotype balance across folds
- LOGO-CV: **Average $R^2 = -0.37$**, showing poor generalization to unseen genotypes
- Some genotypes (BESC-1025, KLND) showed LOGO $R^2 > 0$; others (GW-9842) were consistently negative.

Interpretation:
→ Plant genetics heavily influence predictability. The model learns general growth rules but fails on genotypes with atypical patterns.

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3.3 Biological Meaning & Implications

Why These Results Make Sense

- Early pigment content is tied to energy capture, stress response, and leaf health, which logically links to long-term vigor.
- Early structural growth reflects how quickly a plant establishes biomass, determining future capacity.

Practical Implications

- Early selection: choose high-performing genotypes without waiting 35 days.
- Breeding insights: identify genotypes with strong early indicators.
- Automation potential: spectral cameras + early imaging could replace long monitoring cycles.

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4. Summary

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- Early spectral signals accurately identified high-performing plants, with pigment-based features.
- Early growth patterns (0–14 days) reliably predicted final biomass, with linear models giving the best performance.
- Validation strategies ensured true biological generalization; results were consistent across plants but varied across genotypes.
- Together, the models show that early plant traits can be used for rapid screening, selection, and phenotyping optimization.

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